Plot a Taxonomic Trait Comparison

Description

Builds a 'ggplot2' $\log_{10}$-scale strip plot comparing the median trait value of a focal taxon against taxa in the selected taxonomic group. Grey jittered dots represent group members that meet the minimum record threshold; a firebrick point marks the focal taxon.

Usage

```
plot_taxonomic_trait_comparison(
  data_taxonomic_trait_summary,
  data_focal_trait_summary,
  focal_taxon,
  trait_domain,
  minimum_records = 5L,
  graphical_options,
  verbose = TRUE
)
```

Arguments

`data_taxonomic_trait_summary`	A tibble of per-taxon summary statistics returned by [summarise_taxonomic_group_traits()]. Must contain 'taxon_name', 'n_records', and 'median'.
`data_focal_trait_summary`	A single-row tibble of per-group QC statistics for the focal taxon, as produced by [write_trait_quality_control_report()]. Must contain a numeric column 'median' used to place the firebrick reference point.
`focal_taxon`	Character scalar. Name of the focal taxon. Used in the plot title.
`trait_domain`	Character scalar. Trait domain being inspected. Used in axis labels.
`minimum_records`	Positive integer scalar. Minimum number of records a taxon must have to appear in the plot. Defaults to '5L'.

graphical_options	Named list with elements 'width', 'height', 'units', 'dpi', and 'bg', as returned by 'load_active_config_value("graphical")'. Passed to 'ggview::canvas()'.
verbose	Logical. If 'TRUE' (default), the number of taxa shown in the plot after the 'minimum_records' filter is reported via 'cli::cli_inform()'.

Details

All taxa with 'n_records >= minimum_records' are shown as grey jittered points on a $\log_{10}$ x-axis. The focal taxon's median from 'data_focal_trait_summary' is overlaid as a larger firebrick point. When no group members meet the threshold the grey layer is empty and only the focal taxon point is drawn.

Plot layer order follows project conventions: 'ggplot2::ggplot()' -> scales -> labels -> theme -> 'ggview::canvas()' -> geoms.

Value

A 'ggplot2' object. The plot is not printed; call 'print()' on the return value to display it.

See Also

[summarise_taxonomic_group_traits()], [plot_focal_trait_distribution()], [write_trait_quality_control_report()]

Examples

```
## Not run:

data_taxonomic_trait_summary <-

  summarise_taxonomic_group_traits(

    data_trait_records = data_traits_raw,

    data_taxon_classification = data_classification,

    focal_taxon = "Anacyclus clavatus",

    trait_domain = "Leaf Area"

  )


data_summary <-

  data_qc_report |>

  dplyr::filter(

    taxon_name == "Anacyclus clavatus",

    trait_domain_name == "Leaf Area"
```

```
  )

graphical_options <-
  load_active_config_value("graphical")

p <-
  plot_taxonomic_trait_comparison(
    data_taxonomic_trait_summary = data_taxonomic_trait_summary,
    data_focal_trait_summary = data_summary,
    focal_taxon = "Anacyclus clavatus",
    trait_domain = "Leaf Area",
    minimum_records = 5L,
    graphical_options = graphical_options
  )

base::print(p)

## End(Not run)
```